基于半监督自学习的电池寿命预测

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摘要—对于锂离子电池的准确和可靠的退化与寿命预测是智能预测和健康管理的核心挑战。本文提出了一种基于半监督自学习的电池寿命预测新方法。首先，从部分容量-电压曲线上提取三个健康指标（HIs）。其次，使用源域中三个随机选择的电池的数据构建容量估计模型和寿命预测模型。然后，使用HIs重建历史容量，为寿命模型的自我训练提供伪值。最后，使用自我训练的寿命模型预测未来退化。还包括不确定性表达式，以提供未来容量的概率预测。验证中考虑了不同的应用场景。对于不同老化曲线的电池，仅使用三个已知检查点，寿命预测误差小于23个循环。不同电池类型的预测表明，对于长寿命电池，误差小于50个循环，相对误差小于4.1%；对于短寿命电池，误差小于20个循环，相对误差小于5.21%。本文为当现实世界中的标记容量有限时，指导了寿命预测的适当解决方案。

索引词—退化重构，寿命预测，锂离子电池，概率预测，半监督学习。

I. 引言

寿命预测对于锂离子电池近年来吸引了极大的研究兴趣，因为准确的预测对于智能预测和健康管理 [1], [2]至关重要。寿命或剩余使用寿命（RUL）被定义为，例如，从当前时间到电池达到寿命终结（EOL），通常设定为20%的容量衰减 [3]。寿命预测方法可分为基于模型、数据驱动和混合 [4]。

经验半经验模型是基于模型的两种类别之一，它们被广泛用于通过温度和充电状态等影响因素拟合容量退化曲线。然后，根据拟合曲线 [5]预测其他电池的寿命。另一种基于模型的方法是建立电化学模型来模拟电池的老化 [6]。基于模型方法的主要缺陷是它们的泛化能力差。数据驱动方法将电池视为“黑箱”，仅研究数据本身以映射特征与预测之间的关系 [7]。在电池寿命预测方面，两种子类别被广泛使用，即寿命终结预测和退化曲线预测。对于第一种，特征通过手动[8] 或深度学习 [9], 自动提取，以拟合与电池寿命的回归关系。通过使用新提取的特征的回归模型，可以获得测试电池的寿命。然而，这种方法需要大量数据进行特征工程研究和模型训练。另一个问题是仅预测EOL点。然而，预测退化趋势以更好地理解电池如何老化并失去性能也很重要。因此，第二种子类别开发了容量衰减曲线的序列变化模型[10]。然后使用滚动预测方法预测未来的退化曲线，直到达到EOL点 [11]。主要优点是预测了未来的退化，这估计了退化率和拐点以支持早期维护 [12]。然而，主要挑战是难以正确外推电池退化。混合方法的主要目的是通过集成基于模型的方法和数据驱动方法或不同种类的数据驱动方法来提高预测精度 [13], [14]。混合方法严重依赖于数据驱动方法的有效性，其中滚动预测方法比EOL点预测具有更大的研究和应用意义。因此，提高数据驱动方法的准确性和可靠性是先进混合寿命预测方法的关键挑战和基础。

许多数据驱动算法被用于拟合电池容量序列的非线性退化。支持向量机 [15], [16] 和高斯过程回归 [1],[17] 是两种广泛使用的类型，但其核函数严重

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Semi-Supervised Self-Learning-Based Lifetime Prediction for Batteries

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***Abstract*—Accurate and reliable degradation and lifetime prediction for lithium-ion batteries is the main challenge for smart prognostic and health management. This article proposes a novel semi-supervised self-learning method for battery lifetime prediction. First, three health indicators (HIs) are extracted from the partial capacity-voltage curve. Second, the capacity estimation model and lifetime prediction model are built using data from three randomly selected batteries in the source domain. Then, the HIs are used to reconstruct the historical capacities to provide pseudo values for self-training of the lifetime model. Finally, the self-trained lifetime model is used to predict future degradation. The uncertainty expression is also included to provide the probabilistic prediction of future capacities. Different application scenarios are considered in the verification. The mean lifetime prediction error is less than 23 cycles with only three known checkpoints for batteries aging under different profiles. Predictions for different battery types show that the errors are less than 50 cycles with relative errors less than 4.1% for long lifespan batteries, and less than 20 cycles with relative errors less than 5.21% for short lifespan batteries. This article guides proper solutions for lifetime prediction when the labeled capacities in the real world are limited.**

***Index Terms*—Degradation reconstruction, lifetime prediction, lithium-ion battery, probabilistic prediction, semisupervised learning.**

I. Introduction

LIFETIME prediction for lithium-ion batteries attracts great research interest recently since the accurate prediction is vital for smart prognostic and health management [1], [2]. Lifetime or remaining useful life (RUL) is defined, for example, as the remaining running cycles from the current time to the battery reaching its end of life (EOL), which is generally set as 20% capacity attenuation [3]. Methods for lifetime prediction can be divided into model-based, data-driven, and hybrid [4].

Empirical and semiempirical models, which are two categories in model-based, are widely used to fit the capacity degradation curve by some influencing factors such as temperature and state of charge. Then, the lifetime of other batteries is predicted according to the fitted curve [5]. Another model-based method is to build an electrochemical model to simulate the aging of batteries [6]. The main deficiency of model-based methods is their poor generalization ability. Data-driven method treats the battery as a “black box,” and only the data itself is studied to map the relationship between features and predictions [7]. As for battery lifetime prediction, two subcategories are widely used, i.e., the EOL prediction and degradation curve prediction. For the first one of them, the features are extracted manually [8] or automatically by deep learning [9], to fit the regression relationship with the battery lifetime. The lifetime of testing batteries can be obtained by using the regression model with the newly extracted features. However, this kind of method needs a large amount of data for feature engineering research and model training. Another problem is that only the point of EOL is predicted. However, the degradation trend should also be predicted to better understand how the battery ages and loses performance. Thus, the second kind of subcategory develops the sequence variation model of the capacity attenuation curve [10]. The rolling prediction method is then used to predict the future degradation curve until the EOL point is reached [11]. The main advantage is that future degradation is predicted which estimated the degradation rate and knee points to support early maintenance [12]. However, the main challenge is that it is difficult to correctly extrapolate the battery degradation. The main purpose of the hybrid method is to improve prediction accuracy by integrating model-based methods and data-driven methods or different kinds of data-driven methods [13], [14]. Hybrid methods rely heavily on the effectiveness of data-driven methods, where the rolling prediction method has more research and application significance than EOL point prediction. Therefore, improving the accuracy and reliability of the data-driven methods is the key challenge and basis for advanced hybrid lifetime prediction methods.

Many types of data-driven algorithms are used to fit the nonlinear degradation of the battery capacity sequence. Support vector machine [15], [16] and Gaussian process regression [1], [17] are two widely used types, but the kernel functions heavily

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许多数据驱动算法被用于拟合电池容量序列的非线性退化。支持向量机 [18] 和高斯过程回归 [19] [20], [21] 是两种广泛使用的类型，但核函数严重影响预测结果，而难以选择一种适用于所有容量曲线的核函数。基于神经网络 (NN) 的方法是另一个可以用于拟合容量退化曲线 [18] 的大家族。它可以通过神经元中权重和偏置的特定组合来拟合任何种类的非线性关系，这使得它成为一种可行的 [19] 使用方法。用于电池寿命预测的流行神经网络包括基本神经网络、长短期记忆 (LSTM) 神经网络、卷积神经网络 (CNN) 和门控循环单元神经网络 [20], [21]。其中，LSTM 在历史信息利用方面具有优势，这有利于对序列变化的参数进行预测，例如电池容量的情况。然而，不同电池之间的不同模式使得预测具有不同老化曲线的其他电池的退化变得困难。因此，提出了迁移学习 (TL) 来提高寿命预测精度 [22]。基于神经网络的预测的另一个优点是，基于模型的迁移学习策略可以方便地集成到再训练过程中，这有助于提高预测精度和模型泛化能力。迁移学习的主要目的是利用先前训练模型获得的知识，同时通过仅重新训练某些选定参数来整合测试信息，从而加速训练过程并提高模型精度 [23]。然而，容量退化曲线预测的主要问题在于实际应用中模型再训练的连续变化容量值不足。因此，容量估计模型可用于使用部分过程数据 [24] 来估计容量。已发表论文的另一个局限性在于，迁移学习仅用于预测在相似工作条件下相同电池类型的退化。

本文提出了一种基于半监督自学习（SSSL）策略的新型电池容量退化预测方法，该方法通过历史退化重构和基于少量不连续检查点的寿命模型自训练来实现。主要贡献总结如下。

1) 提出了一种自训练过程，用于利用伪容量改进寿命预测模型。2) 仅需少量不连续检查点即可进行连续容量退化预测。3) 从部分曲线中提取历史信息（HIs）以重构历史容量。4) 所提出的SSSL预测方法具有很好的泛化能力，能够预测不同电池类型或/和不同老化条件下具有不确定性的未来退化趋势。

本文其余部分的结构如下：本文使用的数据集在第二节中介绍。然后，第三节描述了所提出的电池历史容量曲线重构和寿命预测方法。接着，第四节展示并讨论了结果。最后，第五节总结全文。

II. 数据描述

本文使用的数据集包含来自[25], [26] 的公开数据以及作者生成的实验数据。

表 I数据集的详细信息

Battery data set	Geometry	Chemistry	Nominal capacity
SD	Cylindrical	LFP/C	1.1 Ah
TD1	Cylindrical	LFP/C	1.1 Ah
TD2	Prismatic	LFP/C	110 Ah
TD3	Prismatic	LFP/C	100 Ah

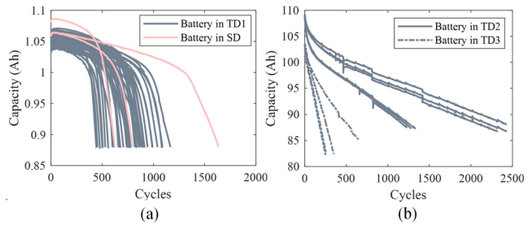


图 1. 电池在 (a) 源域和 TD1 以及 (b) TD2 和 TD3 中的容量衰减曲线。

详细信息参见表 I。数据集分为源域（SD），其中包含从 [24] 随机选出的三种具有不同寿命范围的电池，以及目标域（TD）。在TD中，包含三个数据集。一个是来自 [26] 中介绍的公开数据集，该数据集与SD中的电池类型相同，但具有快速充电协议（命名为TD1）。SD和TD1中电池的容量衰减曲线显示在图 1(a)。由于电池化学和格式相同，且具有较大但相似的电流负载，它们的退化模式相似。然而，图 1(a) 仍然显示不同电池单元之间存在不同的老化速率。原因可能是负载曲线的电流率存在差异，以及不同电池单元存在制造不一致性 [27]。

另外两个实验数据集也包含在TD中。第一个的额定容量为110 Ah（命名为TD2），第二个的额定容量为100 Ah（命名为TD3）。它们来自不同的商业公司，在寿命范围上存在较大差异，如图1(b)所示。这两个数据集的电池格式和工作曲线与SD和TD1中的不同，尽管化学体系相同。TD2和TD3中的电池为方形，而SD和TD1中的为圆柱形。TD2和TD3的负载电流率远小于SD和TD1，充电模式也不同。这些差异使得TD2和TD3中的电池退化模式与SD和TD1中的差异很大。TD2和TD3中的退化曲线形状与SD和TD1中的不同。在TD2和TD3中，电池在早期阶段表现出快速退化率，而在后期阶段存在缓慢且近似线性的退化率，这与SD和TD1中的情况相反。这十个电池（TD2中的电池单元1—6，TD3中的电池单元7–10）在恒流恒压（CCCV）充电和恒流放电条件下进行老化。每个电池的实验详细设置信息在表 II中给出。电池单元1–7

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influence the predictions, and it is hard to choose one kernel function that suits all capacity curves. The neural network (NN) based method is another big family that can be used to fit the capacity degradation curve [18]. It can fit any kind of nonlinear relationship via a specific combination of the weights and biases in the neurons, which makes it a feasible approach to use [19]. Popular NN for battery lifetime prediction includes basic NN, long-short term memory (LSTM) NN, convolutional NN (CNN), and gated recurrent unit NN [20], [21]. Among them, LSTM has superiority in historical information utilization, which benefits the prediction for sequence variated parameters, such as the case of the battery capacity. However, the different patterns among different batteries make it difficult to predict the degradation of other batteries that have different aging curves. Therefore, transfer learning (TL) is proposed to improve the lifetime prediction accuracy [22]. Another advantage of NN-based prediction is that the model-based TL strategy can be conveniently integrated into the retraining process, which helps improve the accuracy of prediction and model generalization ability. The main purpose of TL is to use the knowledge gained from the previously trained model while integrating the testing information by only retraining some chosen parameters, which accelerates the training process and improves the model accuracy [23]. However, the main problem for capacity degradation curve prediction is that there are not enough continuously changed capacity values for model retraining in real applications. Therefore, the capacity estimation model could be used to estimate the capacity with partial process data [24]. Another limitation of published papers is that the TL is only implemented to predict the degradation of the same battery types under similar working conditions.

This article proposes a novel method for battery capacity degradation prediction based on semi-supervised self-learning (SSSL) strategy, which is realized by historical degradation reconstruction and lifetime model self-training based on a few discontinuous checkpoints. The main contributions are summarized as follows.

1) A self-training process is proposed to improve the lifetime prediction model using pseudocapacities.
2) Only a few discontinuous checkpoints are needed for continuous capacity degradations prediction.
3) The HIs are extracted from the partial curves to reconstruct the historical capacities.
4) The proposed SSSL prediction method has a great generalization to predict the future degradation trend of different battery types or/and different aging conditions with uncertainties.

The remainder of the article is structured as follows; the datasets used in this article are introduced in Section II. Then, the proposed battery historical capacity curve reconstruction and lifetime prediction method are described in Section III. Followed that, the results are presented and discussed in Section IV. Finally, Section V concludes this article.

II. Data Description

The datasets used in this article contain both public data from [25], [26] and experimental data generated by the authors. The detailed information is given in Table I. The datasets are divided into the source domain (SD), which contains three randomly selected batteries from [24] with different life ranges, and the target domain (TD). In the TD, three datasets are included. One is from the public dataset presented in [26] that has the same battery types as those in SD but has fast charging protocols (named TD1). The capacity fade curves of the batteries in the SD and TD1 are shown in Fig. 1(a). They have similar degradation patterns due to the same battery chemistry and format with large but similar current loadings. However, Fig. 1(a) still shows that different aging rates exist among different battery cells. The reason could be that the current rates of the loading profiles have differences, and manufactural inconsistency exists among different battery cells [27].

Another two experimental datasets are also included in the TD. The first one has a nominal capacity of 110 Ah (named TD2), and the second has a nominal capacity of 100 Ah (named TD3). They are from different commercial companies and have large differences in the life ranges as shown in Fig. 1(b). These two datasets have different battery formats and working profiles from those in SD and TD1 although the chemistries are the same. The batteries in TD2 and TD3 are prismatic while those in SD and TD1 are cylindrical. The loading current rates for TD2 and TD3 are much less than those for SD and TD1, and the charging modes are also different. These differences make the batteries in TD2 and TD3 have quite different degradation patterns to those in SD and TD1. The shapes of the degradation curves in TD2 and TD3 are different from those in SD and TD1. In TD2 and TD3, batteries show a fast degradation rate in the early stage while a slow and nearly linear degradation rate exists in the later stage, which is opposite to that in SD and TD1. These ten batteries (cells 1—6 in TD2, and cells 7–10 in TD3) are aged under constant current constant voltage (CCCV) charging and CC discharging. The detailed setting information of the experiment for each cell is given in Table II. Cells 1–7 are

TABLE I
Detailed Information of the Datasets

Battery data set	Geometry	Chemistry	Nominal capacity
SD	Cylindrical	LFP/C	1.1 Ah
TD1	Cylindrical	LFP/C	1.1 Ah
TD2	Prismatic	LFP/C	110 Ah
TD3	Prismatic	LFP/C	100 Ah

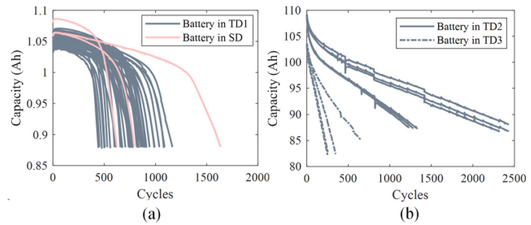


Fig. 1. Capacity fade curve of batteries in (a) SD and TD1, and (b) TD2 and TD3.

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表 IITD2和TD3中的电池实验

Battery cell	Charge-discharge rate	Environment temperature	Capacity range	Cycles
Cell 1	1C-1C	55 °C	87.32-109.15	1282
Cell 2	1C-1C	55 °C	87.46-109.30	1231
Cell 3	1C-1C	55 °C	87.26-109.06	1334
Cell 4	1C-1C	35 °C	88.13-109.07	2431
Cell 5	1C-1C	35 °C	86.83-108.42	2313
Cell 6	1C-1C	35 °C	86.81-107.97	2431
Cell 7	1C-1C	25 °C	82.45-103.04	346
Cell 8	0.5C-0.5C	35 °C	85.23-103.08	645
Cell 9	0.5C-0.5C	55 °C	82.28-102.81	250
Cell 10	0.5C-0.5C	55 °C	82.77-103.44	254

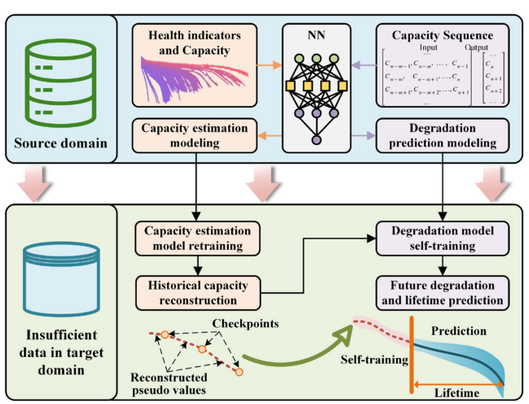


图2。基于SSSL的电池寿命预测总体框架。

以1C电流率老化，电池单元8–10以0.5C电流率老化。电池单元1–3和电池单元9、10的温度为55 °C，电池单元4–6和电池单元8的温度为35 °C，而电池单元5的温度为25 °C。如可观察到，TD2和TD3中的电池在不同条件下进行老化，这导致它们具有不同的退化模式和循环寿命。因此，本文使用的数据集包含了在不同工作条件和不同电池格式/几何形状下的预测场景。

III. 预测方法

基于SSSL的电池寿命预测的整体框架如图2所示。数据集被分为一个源域和一个标签数据不足的目标域。具体来说，将在源域中训练两个基于神经网络的模型，一个用于基于历史信息的容量估计模型，另一个用于容量退化模型。这两个基础模型使用大量数据进行离线训练，以确保模型训练充分。然后，在只有稀疏和有限标签数据的目标域中，首先重新训练容量估计模型，以仅考虑有限数量的检查点重建历史容量退化曲线。历史容量由容量估计模型使用所提出的历史信息重建。随后，退化模型使用估计的伪容量值进行自训练。这里，训练与自训练的主要区别在于使用真实标签还是伪标签。重新训练和自训练过程是在可用的检查点下在线进行的。由于仅使用有限数据，因此可以快速训练，并且基础模型有助于加速收敛速度。因此，模型构建过程包含离线校准和在线调整，以提高估计和预测精度以及模型泛化能力。最后，预测未来退化轨迹，直到达到电池的EOL，以进行电池寿命预测。在退化预测模型中，预测的容量用于迭代预测。与传统的基于LSTM的方法相比，所提出的基于SSSL的方法的主要优势体现在以下三个方面。

训练与自训练的区别在于是否使用真实标签或伪标签。重新训练和自训练过程是在可用的检查点下在线进行的。由于仅使用有限数据，因此可以快速训练，并且基础模型有助于加速收敛速度。因此，模型构建过程包含离线校准和在线调整，以提高估计和预测精度以及模型泛化能力。最后，预测未来退化轨迹，直到达到电池的EOL，以进行电池寿命预测。在退化预测模型中，预测的容量用于迭代预测。与传统的基于LSTM的方法相比，所提出的基于SSSL的方法的主要优势主要体现在以下三个方面。

1) 使用稀疏且有限的标记检查点重新训练容量估计模型，同时未标记的历史数据可以被很好地重建。

2) 估计容量的伪值用于退化预测模型的自训练，以提高未来退化预测的准确性。因此，过程数据（电压、电流和时间）和标记容量都被很好地利用，以提高预测模型的准确性。

3) 概率回归被集成以提供预测容量曲线的置信区间（CI）。在以下小节中，将详细描述建模和训练过程。

A. 容量重建

伪容量重建的主要过程包括历史信息提取、容量估计模型训练，以及使用有限数量的检查点进行再训练。

1) 健康指标提取：大多数实际条件下的不完整充放电需要间接容量估计来支持电池管理，这可以通过基于特征的数据驱动方法实现。众所周知，电池容量-电压（Q–V）曲线中隐藏着丰富的老化信息，这些曲线随着老化呈现规律性变化[19]。本文使用部分Q–V 进行历史信息提取。电压序列从最低电压到最高电压均等分为1000段。本文选择2.85至3.25 V的电压序列。每个循环的电压序列表示为 [$\boldsymbol{V}_m$，$\boldsymbol{V}_{m+1}$，…，$\boldsymbol{V}_{n-1}$，$\boldsymbol{V}_n$]。计算每个电压段之间的充电容量（Q）值，形成Q 序列，表示为 [$\boldsymbol{Q}_m$，$\boldsymbol{Q}_{m+1}$，…，$\boldsymbol{Q}_{n-1}$，$\boldsymbol{Q}_n$]。最后，从Q 序列中提取三个历史信息。此处，包括标准差、香农熵和第一主成分的三个历史信息被计算。标准差和香农熵通过每个循环的Q 序列的对应值计算。第一主成分通过Q矩阵的降维提取，该矩阵由不同循环的Q序列形成。众所周知，Q–V曲线随老化呈现规律性变化。标准差可以描述这种变化的程度。香农熵是反映序列中信息量的良好指标。

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TABLE II
EXPERIMENT OF BATTERIES IN TD2 AND TD3

Battery cell	Charge-discharge rate	Environment temperature	Capacity range	Cycles
Cell 1	1C-1C	55 °C	87.32-109.15	1282
Cell 2	1C-1C	55 °C	87.46-109.30	1231
Cell 3	1C-1C	55 °C	87.26-109.06	1334
Cell 4	1C-1C	35 °C	88.13-109.07	2431
Cell 5	1C-1C	35 °C	86.83-108.42	2313
Cell 6	1C-1C	35 °C	86.81-107.97	2431
Cell 7	1C-1C	25 °C	82.45-103.04	346
Cell 8	0.5C-0.5C	35 °C	85.23-103.08	645
Cell 9	0.5C-0.5C	55 °C	82.28-102.81	250
Cell 10	0.5C-0.5C	55 °C	82.77-103.44	254

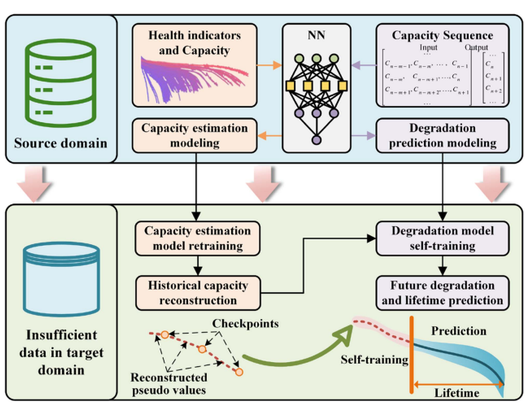


Fig. 2. Overall framework for SSSL-based battery lifetime prediction.

aged with a current rate of 1C, and cells 8–10 are aged at 0.5C. The temperature is 55 °C for cells 1–3 and cells 9 and 10, 35 °C for cells 4–6 and cell 8, while 25 °C for cell 5. As one can observe, the batteries in TD2 and TD3 are aged under different conditions, which causes them to have different degradation patterns and cycle lives. Therefore, the datasets used in this article contain scenarios for predictions under different working conditions and different battery formats/geometry.

III. PREDICTION METHOD

The overall framework of the SSSL-based battery lifetime prediction is shown in Fig. 2. The datasets are divided into a SD and a TD with insufficient labeled data. Specifically, two NNs-based models will be trained in the SD, one for the HIs-based capacity estimation model and the other for the capacity degradation model. These two base models are trained offline using ample data to make sure the models are well trained. Then, in the TD, which has only sparse and limited labeled data, the capacity estimation model is first retrained to reconstruct the historical degradation curve of capacities considering only a limited number of checkpoints. The historical capacities are reconstructed by the capacity estimation model using the proposed HIs. Following that, the degradation model is self-trained using the estimated pseudo capacity values. Here, the main difference between training and self-training is whether real labels or pseudolabels are used. The retraining and self-training processes are conducted online with the available checkpoints. It could be trained fast since only limited data are used, and the base model helps accelerate the convergence speed. Therefore, the model construction process contains offline calibration and online adjustment to improve the estimation and prediction accuracy as well as the model generalization. Finally, the future degradation trajectory is predicted until reaches the EOL for the battery lifetime prediction. In the degradation prediction model, the predicted capacities are used for the iterative prediction. The advantages of the proposed SSSL-based method compared to the conventional LSTM-based method mainly exist in the following three aspects.

1) The sparse and limited labeled checkpoints are used to retrain the capacity estimation model, while the unlabeled historical data could be well reconstructed.

2) The pseudovalues of the estimated capacities are utilized for the self-training of the degradation prediction model to improve the accuracy of the future degradation prediction. Therefore, both the process data (voltage, current, and time) and labeled capacities are well used to improve the accuracy of the prediction models.

3) The probabilistic regression is integrated to provide the confidence interval (CI) of the predicted capacity curve. In the following subsections, the detailed modeling and training process will be described.

A. Capacity Reconstruction

The main process for pseudo capacity reconstruction includes HIs extraction, capacity estimation model training, and retraining with a limited number of checkpoints.

1) Health Indicators Extraction: The incomplete charging and discharging in most real conditions require indirect capacity estimation to support battery management, which can be realized by the feature-based data-driven method. It is known that ample aging information is hidden behind the battery capacity-voltage (Q–V) curves, which show regular changes along with aging [19]. The partial Q–V is used for HIs extraction in this article. The volage sequence is divided equally into 1000 segments from the lowest voltage to the highest voltage. Then, the voltage sequence from 2.85 to 3.25 V is selected in this article. The voltage sequence for each cycle is denoted as [$\boldsymbol{V}_m$, $\boldsymbol{V}_{m+1}$, …, $\boldsymbol{V}_{n-1}$, $\boldsymbol{V}_n$]. The charged capacity (Q) value between each voltage segment is calculated to form the Q sequence, which is denoted as [$\boldsymbol{Q}_m$, $\boldsymbol{Q}_{m+1}$, …, $\boldsymbol{Q}_{n-1}$, $\boldsymbol{Q}_n$]. Finally, three HIs are extracted from the Q sequence. Here, the three HIs including standard deviation, Shannon entropy, and the first principal component are calculated. The standard deviation and Shannon entropy are calculated by the corresponding value of the Q sequence from each cycle. The first principal component is extracted by dimension reduction of the Q matrix, which is formed by the Q sequence at different cycles. It is known that the Q–V curve varies regularly with aging. The standard deviation can describe the degree of this variation. The Shannon entropy is a good indicator to reflect the amount of information contained in the sequence.

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表 III三个历史信息（HIs）的平均相关系数

Data sets	Standard deviation	Shannon entropy	First principal component
SD	0.9681	0.9702	0.9622
TD1	0.9578	0.9609	0.9502
TD2	0.9711	0.9910	0.9659
TD3	0.9774	0.9794	0.9654

主成分分析通过多个主成分描述序列的内部关系，其中第一个成分描述Q序列的主要退化趋势。因此，这三个历史信息是描述电池健康状况的良好指标。

为评估三个HIs的有效性，计算了HIs与电池容量之间的皮尔逊相关系数。四个数据集的平均相关系数值给出在表 III。结果表明，所有三个HIs对于所有四个数据集的平均相关系数均大于0.95，这表明HIs与电池容量具有高度相关性。因此，这三个HIs可用于重建历史容量，以用于下一小节中的伪值准备。

2) 伪容量重建： 根据从部分曲线中提取的HIs，可以通过有限数量的检查点来重建历史退化过程中的伪容量，以进行模型重新训练。首先，使用来自源域（SD）的数据对基于NN的容量估计模型进行预训练，其中输入是上述提取的HIs，输出是每个周期的电池容量。NN由一个使用relu激活函数的LSTM层和一个密集层构成。LSTM的基本计算如 [28]中所述进行。

$$f\ (t) = \sigma\left(w_{f1}x\left(t\right) + w_{f2}h\left(t-1\right) + b_f\right) \tag{1}$$

$$i\ (t) = \sigma\left(w_{i1}x\left(t\right) + w_{i2}h\left(t-1\right) + b_i\right) \tag{2}$$

$$\tilde{S}\ (t) = \tanh\left(w_{c1}x\left(t\right) + w_{c2}h\left(t-1\right) + b_c\right) \tag{3}$$

$$S\ (t) = f\left(t\right) \odot S\left(t-1\right) + i\left(t\right) \odot \tilde{S}\left(t\right) \tag{4}$$

$$o\ (t) = \sigma\left(w_{o1}x\left(t\right) + w_{o2}h\left(t-1\right) + b_o\right) \tag{5}$$

$$h\ (t) = o\left(t\right) \odot \tanh\left(S\left(t\right)\right) \tag{6}$$

其中 $x(t)$ 和 $h(t)$ 分别是输入和输出， $S(t)$ 是状态信息， $f(t)$、 $i(t)$ 和 $o(t)$ 分别是由遗忘门、输入门和输出门更新的信息， w 和 b 分别是权重和偏置， σ 和 tanh 是激活函数。

然后，对于测试电池的历史容量重建，只需要少量带有标签数据的检查点即可用于最后密集层的再训练。这里的检查点可以在实际世界的定期维护过程中获得，这使得它非常现实。最后，再训练的模型用于重建检查点之间的历史容量。重建的容量将作为伪值用于下一节中寿命模型的自我训练过程。

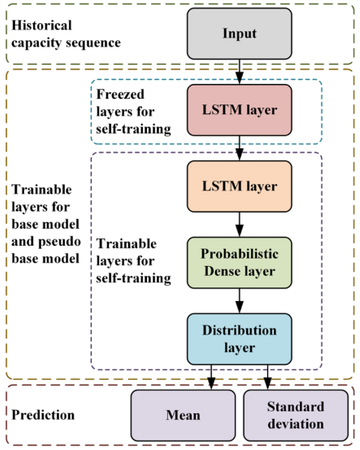


图3。 所提出的寿命模型的框架和训练过程。

B. 寿命预测

本节提出了采用可迁移NN来预测未来容量并附带不确定性的外推方法，该NN在本节中提出。NN的提出框架如图3所示图3。基本训练、迁移基本训练和自我训练过程将在下一节中描述。

1) 寿命建模： 对于基于序列的容量预测，容量首先被重构以形成输入和输出，

$$[C_{i-m}, C_{i-m+1}, \ldots, C_{i-1}, C_i] \tag{7}$$

$$[C_{i+1}, C_{i+2}, \ldots, C_{i+n}] \tag{8}$$

其中 C_i 是第 i 个循环的容量，m 和 n 分别表示输入长度和预测长度。在本文中，m 被设置为 9，n 被设置为 1，这意味着输入和输出的长度分别是 10 和 1。

然后，使用 图3 中所示的神经网络框架进行模型训练。这里，首先添加了两个 LSTM 层。然后，考虑了一个概率密集层。概率密集层的基本功能与密集层相同，但权重和偏置被设置为分布而不是特定值。最后一层是一个 DistributionLambda (Distribution) 层，用于输出预测的分布。使用负对数似然损失函数进行损失最小化 [29]

$$\text{loss}\ (z, y) = -y.\text{log_prob}\left(z\right) \tag{9}$$

其中 y 和 z 分别表示预测分布和真实目标， y.log_prob(z) 表示在特定分布 y 下获取一个实验样本 z 的对数概率（logprob）。寿命模型首先使用源域（SD）中三个电池的数据进行训练。然后，采用SSSL过程来更新模型，用于预测测试域（TD）中电池的寿命，具体将在下一节介绍。

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TABLE III
MEAN CORRELATION COEFFICIENT FOR THE THREE HIS

Data sets	Standard deviation	Shannon entropy	First principal component
SD	0.9681	0.9702	0.9622
TD1	0.9578	0.9609	0.9502
TD2	0.9711	0.9910	0.9659
TD3	0.9774	0.9794	0.9654

The principal component analysis is used to describe the internal relationships of the sequence by several principal components, where the first component describes the main degradation trend of the Q sequence. Therefore, these three HIs are good indicators to describe the health status of batteries indirectly.

To evaluate the effectiveness of the three HIs, Pearson correlation coefficients between the HIs and battery capacities are calculated. The mean correlation coefficient values for the four datasets are given in Table III. The results show that the mean correlation coefficients for all the three HIs regarding all the four datasets is larger than 0.95, which indicates the HIs have high correlations with battery capacities. Therefore, the three HIs could be used to reconstruct the historical capacities for the pseudo values preparation in the next subsection.

2) Pseudocapacity Reconstruction: According to the extracted HIs from the partial curve, the pseudo capacities in the historical degradation process can be reconstructed with a limited number of checkpoints for model retraining. First, the NN-based capacity estimation model is pretrained using the data from the SD, where the inputs are the HIs extracted above and the output is battery capacity for each cycle. The NN is constructed by an LSTM layer with the activation function of *relu* and a dense layer. The basic calculation for the LSTM is performed as presented in [28]

$$f\ (t) = \sigma\left(w_{f1}x\left(t\right) + w_{f2}h\left(t-1\right) + b_f\right) \tag{1}$$

$$i\ (t) = \sigma\left(w_{i1}x\left(t\right) + w_{i2}h\left(t-1\right) + b_i\right) \tag{2}$$

$$\tilde{S}\ (t) = \tanh\left(w_{c1}x\left(t\right) + w_{c2}h\left(t-1\right) + b_c\right) \tag{3}$$

$$S\ (t) = f\left(t\right) \odot S\left(t-1\right) + i\left(t\right) \odot \tilde{S}\left(t\right) \tag{4}$$

$$o\ (t) = \sigma\left(w_{o1}x\left(t\right) + w_{o2}h\left(t-1\right) + b_o\right) \tag{5}$$

$$h\ (t) = o\left(t\right) \odot \tanh\left(S\left(t\right)\right) \tag{6}$$

where $x(t)$ and $h(t)$ are the input and output, respectively, $S(t)$ is the state information, $f(t)$, $i(t)$, and $o(t)$ are the information updated by the forget gate, input gate, and output gate respectively, w and b are the weights and biases, respectively, σ and tanh are the activation functions.

Then, for the historical capacity reconstruction of the tested batteries, only a few checkpoints with labeled data are needed for the retraining of the last dense layer. The checkpoints here could be obtained in the real world during the regular maintenance process, which makes it realistic. Finally, the retrained model is used to reconstruct the historical capacity between the checkpoints. The reconstructed capacities are used as pseudo values for the self-training process of the lifetime model in the following section.

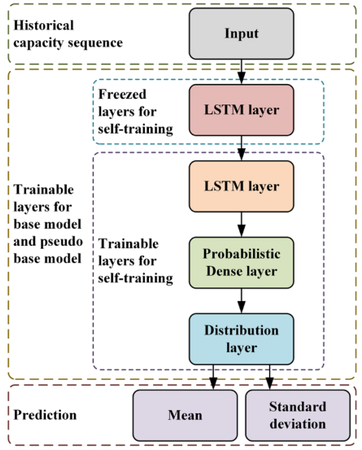


Fig. 3. Framework and training process of the proposed lifetime model.

B. Lifetime Prediction

The extrapolation method is adopted to predict the future capacity with uncertainty based on transferable NN, which is proposed in this section. The proposed framework of the NN is shown in Fig. 3. The basic training, transferred basic training, and self-training processes are described in the following section.

1) Lifetime Modeling: For sequence-based capacity prediction, the capacities are firstly reconstructed to form the inputs and output as

$$[C_{i-m}, C_{i-m+1}, \ldots, C_{i-1}, C_i] \tag{7}$$

$$[C_{i+1}, C_{i+2}, \ldots, C_{i+n}] \tag{8}$$

where C_i is the capacity of ith cycle, m and n represent the input length and predicted length, respectively. In this article, m is set as 9 and n is set as 1, which means the length of input and output are 10 and 1, respectively.

Then, the NN framework shown in Fig. 3 is used for the model training. Here, two LSTM layers are first added. Then, a probabilistic dense layer is considered. The basic function for the probabilistic dense layer is the same as the dense layer, while the weights and biases are set as distributions instead of specific values. The last layer is a DistributionLambda (Distribution) layer to output the predicted distribution. The negative log-likelihood loss function is used for the loss minimization [29]

$$\text{loss}\ (z, y) = -y.\text{log_prob}\left(z\right) \tag{9}$$

where y and z are the predicted distribution and real target respectively, y.log_prob(z) means to get the logarithmic probability (logprob) of one experiment sample (z) under a specific distribution (y). The lifetime model is first trained using the data of the three batteries in the SD. Then, the SSSL process is adopted to update the model for the lifetime prediction of testing batteries in TD, which is introduced in the following section.

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2) 自训练用于预测： 源域中电池的退化曲线与目标域中的不同。如果使用源域训练的寿命模型直接预测目标域中电池的未来退化，预测值将偏离真实值。因此，本文提出了SSSL来提高预测在不同应用下的适应性。

第一种场景是预测与SD中电池相同类型但工作条件不同的电池的寿命。在此，第一个长短期记忆网络层在用SD中电池的数据训练后会被冻结，以保留该类型电池的通用退化特性。第二个长短期记忆网络、概率密集层和分布层设置为可训练，以学习每个测试电池的特定信息。测试电池的重构历史容量曲线用于自训练以改进寿命模型。最后，通过外推预测值来预测未来退化曲线。

第二种场景是预测与SD中电池不同类型的电池的寿命。在这种情况下，从TD中选择一个电池，使用七个检查点（第1个、10%循环、30%循环、50%循环、70%循环、90%循环和最后一个循环）重构整个退化曲线。然后，使用伪容量重新训练整个寿命模型以适应这种电池的退化模式。在此，由SD中电池训练的寿命模型为训练提供参数初始值。模型重新训练后，使用上一段中相同的自训练过程对TD中其他电池的退化进行预测。在这种场景中，使用两个自训练过程来改进用于退化预测的寿命模型。

IV. 结果与讨论

在本节中，使用上述讨论的两个应用场景进行验证。通过分析寿命预测误差以及与真实曲线相比的未来预测容量曲线的准确性来讨论评估结果。本文提出的预测框架基于 Keras 和 TensorFlow Probability 构建。

A. 预测相同电池类型但不同配置

本节讨论了具有相同电池类型但不同工作条件的电池的预测结果。虽然具有相同电池类型的电池具有相似的退化模式，但退化率和形状仍然存在差异，这已在 图 1(a) 中展示。此处使用 SD 中的三个电池进行基础模型训练。然后，在 TD1 中实现 SSSL，以验证寿命预测的有效性。

整个预测容量范围内的寿命预测误差（预测 RUL 与真实 RUL 在起始点之间的不同循环）、均方根误差（RMSE）和平均绝对误差（MAE）显示在 图 4 中。RMSE 和 MAE 是通过整个预测容量计算的，这些容量包括历史重构部分和未来预测部分

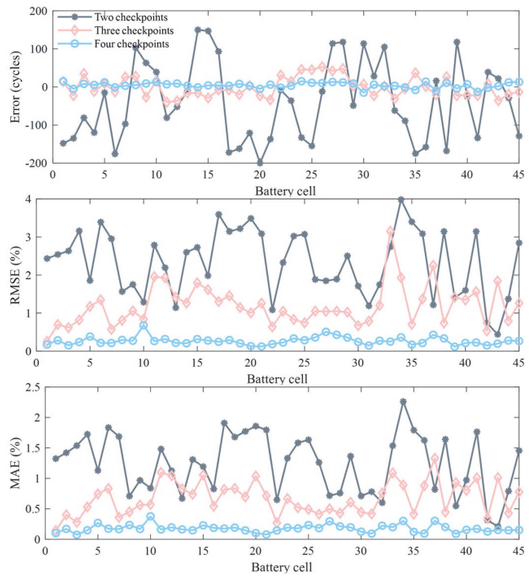


图 4。 TD1 中所有 45 块电池的预测寿命误差、容量均方根误差和平均绝对误差。

表 IVTD1 的预测误差

Checkpoints	Mean absolute error of lifetime	Mean RMSE of capacities	Mean MAE of capacities
2	96.40 cycles	2.33%	1.24%
3	22.68 cycles	1.19%	0.67%
4	7.27 cycles	0.27%	0.18%

部分数据合并。这些 45 个电池单元的平均误差如表 IV 所示。这里的检查点是指测试电池单元的一些已知数据，即整个循环寿命的 10%、30%、50% 和 90% 处的数据。两个检查点表示包含前两个检查点，三个检查点包含前三个，最后一种情况包含所有四个检查点。请注意，在实际应用中，为模型构建获取足够的标记数据既困难又不现实，而稀疏标记数据可以获取（例如，在定期维护中）。因此，本文使用稀疏标记检查点进行再训练和自训练过程，以展示所提出的 SSSL 方法在电池退化曲线和寿命预测方面的改进。图 4 显示，随着用于自训练的检查点数量增加，预测误差显著减少。当有三个检查点可用时，绝对寿命预测误差在 50 个循环内变化。所有 RMSEs 都小于 3.15%，所有 MAEs 都小于 1.4%。尽管使用了三个检查点，但由于电池早期退化非常缓慢，包含的容量范围非常小。因此，结果具有低预测误差，并且对所有电池单元具有良好的泛化能力。当在电池后期使用期间获得另一个检查点时，

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2) Self-Training for Prediction: The degradation curves of the battery in SD are different from those in TDs. If the trained lifetime model by SD is directly used to predict the future degradation of batteries in TDs, the predictions would separate from real values. Therefore, the SSSL is proposed in this article to improve the adaptation of the predictions under different applications.

The first scenario is to predict the lifetime of batteries that have the same battery types, but different working conditions compared with the batteries in SD. Here, the first LSTM layer is freezed after being trained by the data of batteries in SD to retain the general degradation characteristics of this type of battery. The second LSTM, probabilistic Dense layer, and Distribution layer are set trainable to learn specific information about each testing battery. The reconstructed historical capacity curve of the testing battery is used for self-training to improve the lifetime model. Finally, the future degradation curve is predicted by extrapolating the predicted values.

The second scenario is to predict the lifetime of batteries that have different battery types to the batteries in SD. In this condition, one battery in TD is selected to reconstruct the whole degradation curve with seven checkpoints (1st, 10% cycle, 30% cycle, 50% cycle, 70% cycle, 90% cycle, and the last cycle). Then, the whole lifetime model is retrained to adapt to the degradation patterns of this kind of battery using the pseudo capacities. Here, the lifetime model trained by batteries in SD provides the initial values of parameters for training. After the model is retrained, the same process of the self-training in the previous paragraph is used for the degradation prediction of other batteries in TD. In this scenario, two self-training processes are used to improve the lifetime model for degradation prediction.

IV. Results and Discussion

In this section, the two application scenarios discussed above are used for verification. The evaluation is discussed by analyzing both the lifetime prediction error and the accuracy of the future predicted capacity curve compared to the real curve. The prediction framework proposed in this article is built based on Keras and TensorFlow Probability.

A. Prediction for Same Battery Types With Different Profiles

The results of the prediction for batteries that have the same battery types but different working conditions are discussed in this section. Although, the batteries with same battery types have similar degradation patterns, the degradation rates and shapes still show differences, which have been shown in Fig. 1(a). Here the three batteries in SD are used for base model training. Then, the SSSL is implemented in TD1 to demonstrate the effectiveness of the lifetime prediction.

The lifetime predicted errors (different cycles between the predicted RUL and real RUL from start points), root-mean-square error (RMSE), and mean absolute error (MAE) of the whole predicted capacities are shown in Fig. 4. The RMSE and MAE are calculated by the whole predicted capacities, which include the historical reconstructed part and future predicted

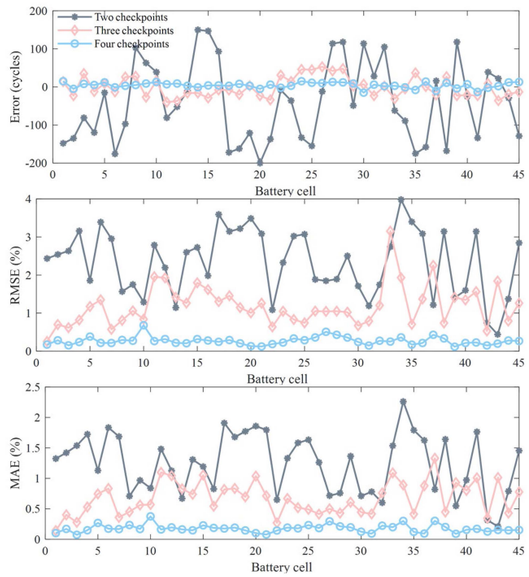


Fig. 4. Predicted lifetime error, capacity RMSE and MAE of all the 45 batteries in TD1.

TABLE IV
Predicted Errors of TD1

Checkpoints	Mean absolute error of lifetime	Mean RMSE of capacities	Mean MAE of capacities
2	96.40 cycles	2.33%	1.24%
3	22.68 cycles	1.19%	0.67%
4	7.27 cycles	0.27%	0.18%

part together. The mean errors of these 45 battery cells are given in Table IV. The checkpoints here mean some known data of the testing battery cell, which are the data at 10%, 30%, 50%, and 90% of the whole cycle life. Two checkpoints mean the first two checkpoints are included, three checkpoints include the first three, and all four checkpoints are included in the last case. Note that it is hard and unrealistic to obtain sufficient labeled data for model construction in practical applications, while sparsely labeled data could be obtained (e.g., in regular maintenance). Therefore, the sparsely labeled checkpoints are used for the retraining and self-training process in this article to demonstrate the improvement of the prosed SSSL method in the battery degradation curve and lifetime prediction. Fig. 4 shows that the prediction errors reduce significantly with the increase of checkpoints used for self-training. When three checkpoints are available, the absolute lifetime prediction errors vary within 50 cycles. All the RMSEs are less than 3.15% and all MAEs are less than 1.4%. Although three checkpoints are used, a very few capacities range is included since the capacity degrades very slowly in the early stages. Therefore, the results have low prediction errors and good generalization for all the cells. When one more checkpoint is obtained during the later battery usage,

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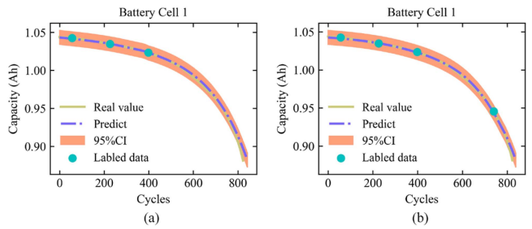


图5。 使用(a)三个检查点和(b)四个检查点对电池单元1的预测结果。

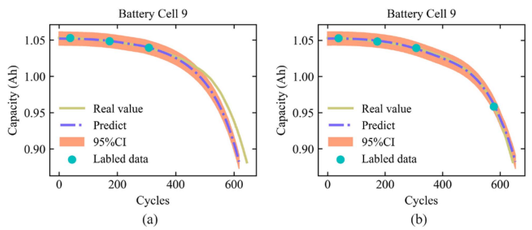


图6。 使用(a)三个检查点和(b)四个检查点对电池单元9的预测结果。

其处于变化更快的阶段，预测误差迅速下降。所有绝对预测寿命误差都小于 15 个循环。所有 RMSEs 和 MAEs 分别小于 0.68% 和 0.38%。

寿命预测和未来容量预测的平均误差给出在 表IV。它表明，当仅使用早期两个检查点时，平均寿命预测误差为96.4周期，当第三个检查点可用时，该误差急剧减少到22.68。当前的均方根误差和平均绝对误差仅为1.19%和0.67%，这意味着预测容量接近实际容量。当获得一个处于快速退化阶段的检查点时，平均寿命预测误差仅为7.27周期，平均均方根误差和平均绝对误差分别为0.27%和0.18%。因此，TD1中所有电池的统计结果表明，提出的方法能够令人满意地预测未来容量退化，并且对不同工作条件和寿命范围具有良好的泛化能力。

TD1 中三个电池的预测结果如图 5–7所示，分别是 TD1 中电池单元 1、9 和 3 的结果。每个图中，子图 (a) 代表由三个可用检查点预测的结果，子图 (b) 是由四个可用检查点获得的结果。这三个电池单元的选择是根据获得三个检查点时的预测结果进行的。图 5(a)代表了预测寿命接近真实寿命的情况（考虑三个检查点）。图 6(a)代表了预测寿命被低估的情况，而图 7(a)代表了预测寿命被高估的情况，这两种情况均考虑了三个检查点。图 5(a)表明，尽管寿命预测准确，但未来容量继续变得更加准确

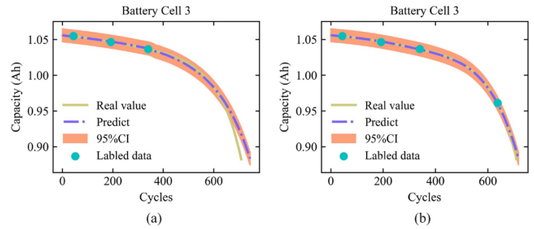


图7。 使用(a)三个检查点和(b)四个检查点对电池单元3的预测结果。

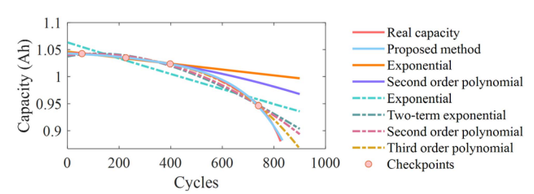


图8。 提出的方 法与曲线拟合方法的比较。

随着一个额外的检查点，如图 5(b)所示。预测容量曲线的 RMSE 和 MAE 分别从 0.26% 和 0.14% 降低到 0.17% 和 0.10%。在图 6(a) 中，预测寿命比真实寿命短，误差为 27 个循环。但是，预测的未来容量曲线仍然与真实曲线显示出相似的退化模式。容量曲线的预测 RMSE 和 MAE 分别为 1.06% 和 0.56%。然而，当多一个位于快速变化过程中的检查点可用时，预测曲线更好地收敛到真实曲线。预测寿命误差减少到 9 个循环，预测容量曲线的 RMSE 和 MAE 分别降低到 0.27% 和 0.17%。对于电池单元 3，在图 7(a)中，预测寿命比真实值长，误差为 35 个循环。同样，预测的未来容量曲线仍然接近真实值，RMSE 和 MAE 分别仅为 0.62% 和 0.15%。当获得第四个检查点，如图7(b)所示，预测变得更好，寿命预测误差仅为 8 个循环。目前整个预测容量曲线的 RMSE 和 MAE 分别仅为 0.15% 和 0.07%。95% CI 覆盖了真实值，以显示预测值的可靠性。

从上述结果可以看出，所提出的方法被证明在仅使用少量检查点的情况下，能够重建历史退化曲线并预测未来轨迹，表现出令人满意的效果。为了进一步评估所提出的预测框架，在已知检查点的曲线拟合方法中进行了比较，结果展示在图8中。这些结果来自电池单元1。实线表示真实值，以及使用三个检查点获得的预测值。由于经验模型的未知参数应少于已知检查点，因此使用了指数函数和二次多项式函数。结果表明，曲线

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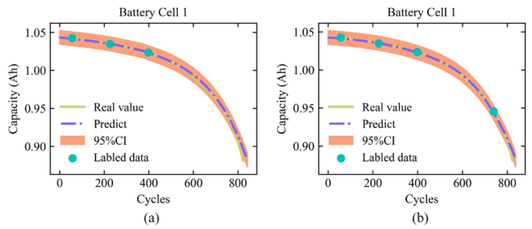


Fig. 5. Prediction results for battery cell 1 using (a) three checkpoints and (b) four checkpoints.

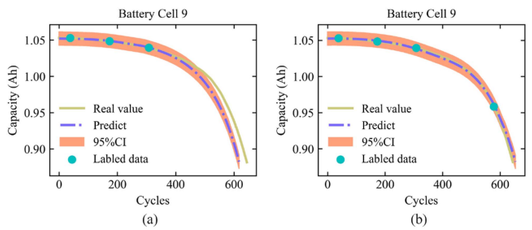


Fig. 6. Prediction results for battery cell 9 using (a) three checkpoints and (b) four checkpoints.

which is in the faster changing stage, the predicted errors decline rapidly. All the absolute predicted lifetime errors are less than 15 cycles. All the RMSEs and MAEs are less than 0.68% and 0.38% respectively.

The mean errors of the lifetime prediction and future capacity prediction are given in Table IV. It shows that when only two checkpoints in early stages are used, the mean lifetime prediction error is 96.4 cycles, which is reduced sharply to 22.68 when the third checkpoint is available. The mean RMSE and MAE currently are only 1.19% and 0.67%, which means the predicted capacities are close to the real capacities. When one more checkpoint that is in the fast-degrading stage is obtained, the mean lifetime prediction error is only 7.27 cycles, and the mean RMSE and MAE are only 0.27% and 0.18%, respectively. Therefore, the statistical results of all the batteries in TD1 illustrate that the proposed method predicts the future capacity degradation satisfactorily and has good generalization for different working conditions and lifetime ranges.

The prediction results of three batteries in TD1 are shown in Figs. 5–7, which are the results of cells 1, 9, and 3 in TD1 respectively. In each figure, subfigure (a) represents the results predicted by three available checkpoints and subfigure (b) is the results obtained by four available checkpoints. The selection of these three battery cells is performed according to the prediction results when three checkpoints are obtained. Fig. 5(a) is representative of the case when the predicted lifetime is close to the real lifetime considering three checkpoints. Fig. 6(a) is representative of the case when the predicted lifetime is underestimated, while Fig. 7(a) is representative of the case when the predicted lifetime is overestimated, both cases considering three checkpoints. Fig. 5(a) shows that although the lifetime prediction is accurate, the future capacity continues to become more accurate

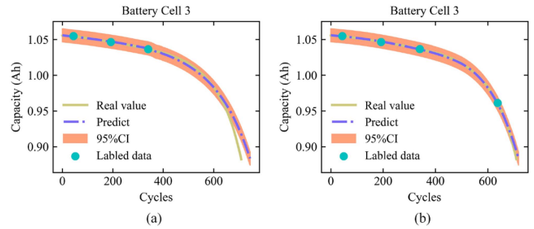


Fig. 7. Prediction results for battery cell 3 using (a) three checkpoints and (b) four checkpoints.

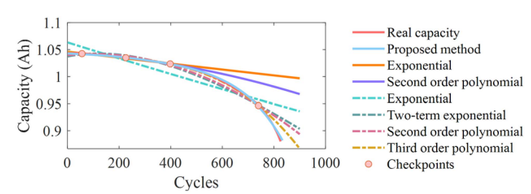


Fig. 8. Comparison between the proposed method and curve fitting methods.

with one more checkpoint, as illustrated in Fig. 5(b). The RMSE and MAE of the predicted capacity curve are reduced to 0.17% and 0.10% from 0.26% and 0.14% respectively. In Fig. 6(a), the predicted lifetime is shorter than the real lifetime, which has an error of 27 cycles. But, the predicted future capacities still show similar degradation patterns to the real curve. The predicted RMSE and MAE of the capacity curve are 1.06% and 0.56%, respectively. However, when one more checkpoint that lies in the fast-changing process is available, the predicted curve converges to the real curve better. The predicted lifetime error is reduced to 9 cycles, and the RMSE and MAE of the predicted capacity curve are reduced to 0.27% and 0.17%, respectively. As for battery cell 3, in Fig. 7(a), the predicted lifetime is longer than the real value with an error of 35 cycles. Similarly, the future predicted capacity curve is still close to the real values with the RMSE and MAE at only 0.62% and 0.15%, respectively. When the fourth checkpoint is obtained as shown in Fig. 7(b), the prediction becomes better, with a lifetime prediction error only of eight cycles. The RMSE and MAE of the whole predicted capacity curve are only 0.15% and 0.07% currently. The 95% CI covers the real values to show the reliability of the predicted values.

From the results presented above, the proposed method is proved to have satisfactory performance to reconstruct the historical degradation curve and predict the future trajectory with only a few checkpoints. To further evaluate the proposed prediction framework, the comparison with curve fitting methods with the known checkpoints is presented in Fig. 8. The results are obtained from battery cell 1. The solid lines represent the real value and the predictions obtained with three checkpoints. Since the unknown parameters of the empirical model should be less than the known checkpoints, the exponential and second-order polynomial functions are used. The results indicate that the curve

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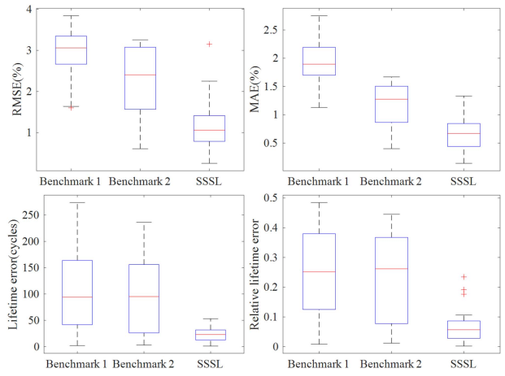


图9。 提出的SSSL方法与两个基准测试的比较。

拟合方法在检查点如此有限的情况下未能预测真实退化曲线。当获得四个检查点时，由虚线表示的拟合曲线与真实曲线仍存在较大偏差。由三次多项式函数获得的最为相似的预测仍然存在52个周期的误差，这远大于所提出的方法在使用仅三个检查点时的预测结果。因此，比较表明，在有限的已知标记数据情况下，所提出的方法对于电池未来容量和寿命预测具有更高的准确性。

为评估提出的SSSL方法在电池寿命预测方面的改进，使用了两个基准测试进行比较。这两个基准测试在容量估计和退化轨迹预测方面采用相同的模型结构。区别在于容量估计模型是否通过稀疏检查点进行再训练，以及退化预测模型是否通过估计的伪容量进行自训练。基准测试1指的是使用基础容量估计模型来估计容量，并使用估计的容量（不进行任何再训练或自训练过程）来预测未来容量衰减的退化模型所获得的结果。基准测试2指的是容量估计模型通过稀疏检查点进行再训练，并使用再训练的模型来估计容量。然后，使用基础退化预测模型根据估计的容量预测退化曲线。而提出的SSSL方法则通过估计的容量对退化预测模型进行自训练以改进其性能，并预测未来退化曲线。因此，这意味着基准测试1没有任何再训练或自训练，基准测试2有再训练但没有自训练，而提出的SSSL方法既有再训练也有自训练。可以看出，基准测试1是用于电池寿命预测的基本长短期记忆网络方法，而基准测试2是用于电池寿命预测的基本长短期记忆网络与Transformer方法。

本次比较使用了三个检查点。容量和寿命预测误差的箱线图显示在图9中。结果表明，当容量估计模型在基准测试2中重新训练时，容量估计误差会减小，与基准测试1相比

表 V两个基准测试与 SSSL 的对比结果

Index	Benchmark 1	Benchmark 2	SSSL
RMSE (%)	2.93%	2.27%	1.19%
MAE (%)	1.92%	1.18%	0.67%
Lifetime error (cycles)	106.42	81.93	22.68
Relative lifetime error (%)	25.09%	20.54%	6.37%
Computational time (s)	21.28	28.06	38.49

要小。然而，所有误差仍然远大于提出的方法中的误差，因为未来退化预测（在基准测试1和2中）不如提出的方法准确。基准测试2的寿命预测与基准测试1相比略有提高。原因可能是尽管退化曲线预测不佳，但预测寿命的最终误差可能差异不大。但是，提出的SSSL方法显著降低了容量和寿命预测误差，这意味着容量估计模型重新训练和退化模型自训练有助于提高整个退化曲线的预测精度。

目标域 1 中 45 个电池的预测容量平均 RMSE 和 MAE 绝对值、预测寿命的平均误差和平均相对误差，以及整个预测过程的计算时间如表 V^{所示}。结果表明了从基准测试 1（可视为传统长短期记忆网络方法）到基准测试 2（可视为长短期记忆网络与 Transformer 结合）再到提出的方法的改进趋势。提出的方法的平均 RMSE 和 MAE 与传统长短期记忆网络方法（基准测试 1）相比分别降低了 59.39% 和 65.10%。寿命预测误差和相对寿命预测误差与传统长短期记忆网络方法相比分别降低了 78.69% 和 74.61%。计算时间由容量估计模型重新训练、退化预测模型自训练、历史容量重构和未来容量预测的时间总和计算得出。45 个电池单元的计算平均时间是表 V^{所示} 的最终计算时间。结果表明，进一步的重新训练和自训练过程不需要过多的额外计算，仅增加了约 17 秒（不到 81%）。考虑到令人满意的精度提升和可接受的计算负担增加，提出的方法在稀疏和有限的标记检查点下对电池退化和寿命预测显示出显著改进。

B. 不同电池类型的预测

上述部分中提出的预测代表了相同电池类型但具有不同老化曲线的应用。在实际应用中，对于具有不同退化模式的不同电池类型的预测也广泛需求。因此，本节评估了在适用场景下使用提出的方法进行的预测。在此应用中，由于退化模式与源域中的电池差异很大，首先使用一个具有七个检查点的电池来重构寿命模型的容量曲线

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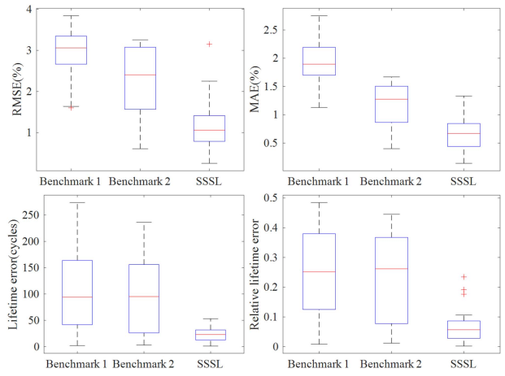


Fig. 9. Comparisons between the proposed method and two benchmarks.

fitting methods failed to predict the real degradation curve with such limited checkpoints. When four checkpoints are obtained, the fitted curves, represented by dot-dash lines, still have large deviations from the real curve. The most similar prediction obtained by the third-order polynomial function still has an error of 52 cycles, which is much larger than the prediction of the proposed method with only three checkpoints. Therefore, the comparison indicates that the proposed method has higher accuracy for battery future capacity and lifetime prediction with limited known labeled data.

To evaluate the improvement of the proposed SSSL for battery lifetime prediction, two benchmarks are used for comparison. The two benchmarks adopt the same model structure for capacity estimation and degradation trajectory prediction. The differences are whether the capacity estimation model is retrained by the sparse checkpoints and the degradation prediction model is self-trained by the estimated pseudo capacities. Benchmark 1 refers to the results obtained by the base capacity estimation model to estimate the capacity and degradation model to predict the future capacity fade using the estimated capacity without any retraining or self-training process. Benchmark 2 refers to that the capacity estimation model is retrained by the sparse checkpoints and the capacity is estimated by the retrained model. Then, the degradation curve is predicted by the estimated capacities using the base degradation prediction model. While the proposed SSSL method conducted self-training to improve the degradation prediction model using the estimated capacities and then predict the future degradation curves. Therefore, it means that benchmark 1 is without any retraining or self-training, benchmark 2 has retraining but no self-training, and the proposed SSSL has both retraining and self-training. It could be seen that benchmark 1 is the basic LSTM method for battery lifetime prediction, and benchmark 2 is the basic LSTM with TL for battery lifetime prediction.

Three checkpoints are used in this comparison. The boxplot for the capacity and lifetime prediction errors is shown in Fig. 9. It shows that the capacity estimation errors are reduced when the capacity estimation model is retrained in benchmark 2 compared

TABLE V
COMPARATIVE RESULTS BETWEEN TWO BENCHMARKS AND THE SSSL

Index	Benchmark 1	Benchmark 2	SSSL
RMSE (%)	2.93%	2.27%	1.19%
MAE (%)	1.92%	1.18%	0.67%
Lifetime error (cycles)	106.42	81.93	22.68
Relative lifetime error (%)	25.09%	20.54%	6.37%
Computational time (s)	21.28	28.06	38.49

to that in benchmark 1. However, all errors are still much larger than those in the proposed method, because the future degradation prediction (in benchmarks 1 and 2) is poorer than for the proposed method. The lifetime prediction of benchmark 2 improves slightly compared to benchmark 1. The reason could be that although the degradation curve is poorly predicted, the final errors of the predicted lifetime may not show much difference. But, all the capacity and lifetime prediction errors are reduced significantly by the proposed SSSL method, which means the capacity estimation model retraining and degradation model self-training helps improving the prediction accuracy for the whole degradation curve.

The absolute values of the mean RMSE and MAE for the predicted capacities, the mean error and mean relative error for the predicted lifetime, and the computational time of the whole prediction process for the 45 batteries in TD1 are given in Table V. The results show the improvement trend from benchmark 1 (which can be seen as the traditional LSTM method) to benchmark 2 (which can be seen as LSTM with TL), and to the proposed method. The mean RMSE and MAE of the proposed method are reduced by 59.39% and 65.10% compared to the traditional LSTM method (benchmark 1). The lifetime prediction error and relative lifetime prediction error are reduced by 78.69% and 74.61% compared to the traditional LSTM method. The computational time is calculated by the time for the whole capacity estimation model retraining, degradation prediction model self-training, historical capacity reconstruction, and future capacity prediction. The mean time for the calculation of the 45 battery cells is the final computational time in Table V. The results show that the further retraining and self-training process does not require too much additional computation, which just increases by about 17s (less than 81%). Considering the satisfactory accuracy improvement and the acceptable computational burden increment, the proposed method shows great improvement in the battery degradation and lifetime predictions with sparse and limited labeled checkpoints.

B. Predictions for Different Battery Types

The predictions presented in the above section represent the applications for the same battery type with different aging profiles. In real applications, the prediction for different battery types that have different degradation patterns is also widely required. Therefore, the prediction under this applicable scenario using the proposed method is evaluated in this section. In this application, since the degradation patterns are quite different from the batteries in SD, one battery with seven checkpoints is first used to reconstruct the capacity curve for lifetime model

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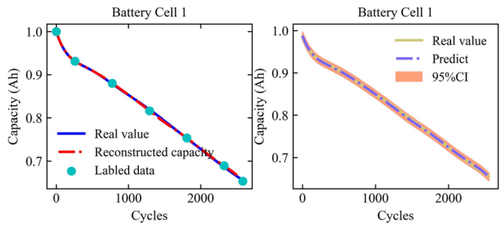


图 10。 容量重建和寿命模型再训练。

进行自训练。然后，按照上述部分中的过程进一步改进自训练模型，以预测这种新型但具有不同老化特征的电池的未来容量和寿命。

1) 目标域 2 中电池的预测： 本节介绍了目标域 2 中电池的预测结果及其评估。首先，使用目标域 2 中的电池单元 1，通过七个检查点来重新训练所有网络，以学习这种电池类型的特定特征。容量重建和寿命预测模型训练结果如图 10 所示。左子图是容量重建结果，右子图是再训练的寿命模型结果。获得的结果表明，仅使用七个检查点就可以很好地重建单个电池的容量衰减曲线。原因是用于容量估计的历史信息与所有四个数据集的电池容量具有高皮尔逊相关系数。模型只需几个检查点即可快速更新并快速收敛。整个容量退化曲线可以很好地重建，为寿命模型再训练提供伪值。然后，使用估计的伪容量重新训练整个退化预测模型。由伪值训练的寿命模型也正确地覆盖了真实的退化曲线，95% 置信区间较小。尽管只使用了少量检查点，但半监督学习策略可以学习新电池类型在新的老化特征下的特定退化模式。因此，再训练的模型可以像前一节一样用于预测相同电池类型但具有不同退化曲线的其他电池的寿命。

电池单元–6的寿命预测结果如图 11 所示。此处仅使用了早期三个检查点（第 100、200 和 300 次循环）。结果表明，尽管寿命范围不同，但这五个电池单元的预测都准确。未来容量正确地遵循了真实值，95% 置信区间成功地收敛于真实容量曲线，这表明所提出的方法能够有效预测不同电池类型和不同老化特征的电池寿命。为了评估首次重新训练电池选择的影响，首先使用电池单元 6 重新训练整个神经网络，并使用三个早期检查点通过持续自训练来预测电池单元 1 的寿命。结果表明，预测也是准确且可靠的。因此，所提出的方法具有良好的泛化能力。

TD2 的寿命预测数值结果如表 VI 所示，以及未来预测容量的误差

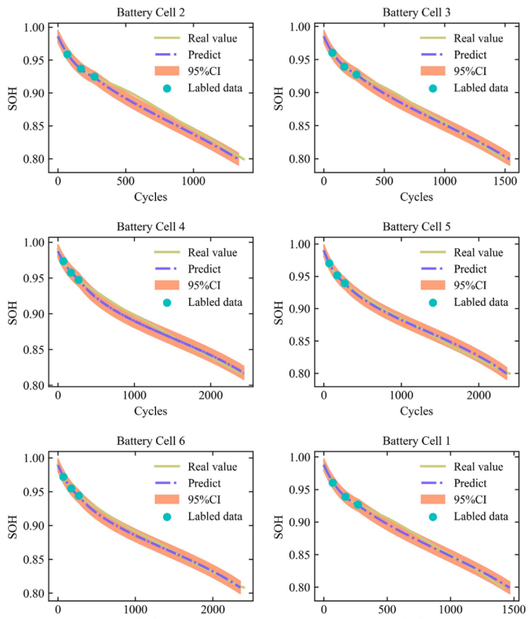


图 11.TD 中的电池寿命预测结果

2.

表 VITD2 寿命预测误差

Battery cell	Real lifetime	Predicted lifetime	Absolute error	Relative error
Cell 1	982	1029	47 cycles	4.08%
Cell 2	931	893	-38 cycles	-3.44%
Cell 3	1034	1082	48 cycles	3.93%
Cell 4	2131	2164	33 cycles	1.55%
Cell 5	2013	1975	-38 cycles	1.78%
Cell 6	2131	2088	-43 cycles	2.02%

表 VIITD2 容量预测误差

Battery cell	RMSE of capacities	MAE of capacities
Cell 1	0.51%	0.41%
Cell 2	0.81%	0.70%
Cell 3	0.53%	0.41%
Cell 4	0.54%	0.45%
Cell 5	0.36%	0.30%
Cell 6	0.44%	0.34%

曲线给出在 表 VII。在 55 °C 下老化的电池剩余寿命约为 1000 次循环，而在 35 °C 下老化的电池剩余寿命超过 2000 次循环。结果表明，这六种电池的预测寿命误差都小于 50 次循环。因此，在 55 °C 下老化且剩余寿命约为 1000 次循环的电池，其相对误差小于 4.1%。此外，在 35 °C 下老化且剩余寿命超过 2000 次循环的电池，其相对误差小于 2.1%。在这种情况下，预测误差足够低，以

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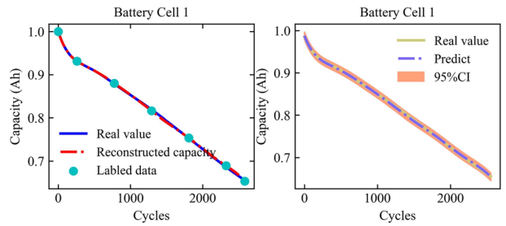


Fig. 10. Capacity reconstruction and lifetime model retraining.

self-training. Then, the self-trained model is further improved following the process in the above section to predict the future capacity and lifetime of this new type but different aging characteristic batteries.

1) Predictions for Batteries in TD2: The predictions for batteries in TD2 are presented and evaluated in this section. First, battery cell 1 in TD2 is used to retrain all the networks with seven checkpoints to learn the specific characteristics of this kind of battery type. The capacity reconstruction and lifetime prediction model training results are shown in Fig. 10. The left subfigure is the capacity reconstruction results and the right subfigure is the lifetime model retrained results. The obtained results show that the capacity fade curve of one battery can be well reconstructed with only seven checkpoints. The reason is that the HIs used for capacity estimation have high Pearson correlation coefficients with battery capacities for all four datasets. The model could be updated fast and converged quickly with only a few checkpoints. The whole capacity degradation curve could be well reconstructed to provide the pseudo values for lifetime model retraining. Then, the estimated pseudo capacities are used to retrain the whole degradation prediction model. The lifetime model trained by the pseudo values also covers the real degradation curve correctly with small 95% CIs. Although only a few checkpoints are used, the specific degradation pattern of the new battery-type under new aging profiles could be learned using the semisupervised learning strategy. Therefore, the retrained model can be used in the same way as in the former section to predict the lifetime of other batteries of the same battery types but with different degradation curves.

The lifetime prediction results for battery cells–6 are shown in Fig. 11. Here, only three checkpoints at the early stage (100th, 200th, and 300th cycle) are used. The results show that the predictions for these five battery cells are accurate despite different life ranges. The future capacities follow the real values correctly and the 95% CIs converge the real capacity curve successfully, indicating the effectiveness of the proposed method to predict the lifetime of different battery types and different aging profiles. To evaluate the influence of selection of the first retrained battery, cell 6 is firstly used to retrain the whole NN and to predict the lifetime of cell 1 by continued self-training using the three early checkpoints. The results show that the prediction is also accurate and reliable. Therefore, the proposed method has good generalization ability.

The numerical results of the lifetime predictions for TD2 are given in Table VI, and the errors of the future predicted capacity

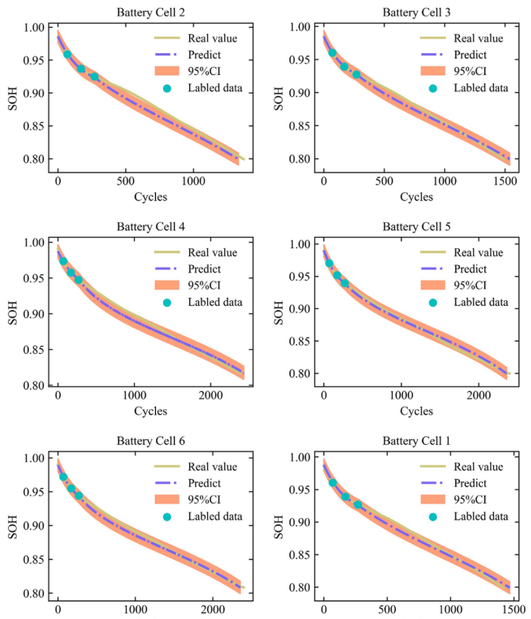


Fig. 11. Lifetime prediction results for batteries in TD2.

TABLE VI
LIFETIME PREDICTED ERRORS OF TD2

Battery cell	Real lifetime	Predicted lifetime	Absolute error	Relative error
Cell 1	982	1029	47 cycles	4.08%
Cell 2	931	893	-38 cycles	-3.44%
Cell 3	1034	1082	48 cycles	3.93%
Cell 4	2131	2164	33 cycles	1.55%
Cell 5	2013	1975	-38 cycles	1.78%
Cell 6	2131	2088	-43 cycles	2.02%

TABLE VII
CAPACITY PREDICTED ERRORS OF TD2

Battery cell	RMSE of capacities	MAE of capacities
Cell 1	0.51%	0.41%
Cell 2	0.81%	0.70%
Cell 3	0.53%	0.41%
Cell 4	0.54%	0.45%
Cell 5	0.36%	0.30%
Cell 6	0.44%	0.34%

curve are given in Table VII. The remaining lifetime of the batteries aged under 55 °C is about 1000 cycles while that of the batteries aging under 35 °C is more than 2000 cycles. The results show that the predicted lifetime errors are less than 50 cycles for all these six batteries. Therefore, the relative errors are less than 4.1% for the batteries aged under 55 °C with about 1000 cycles remaining. Furthermore, the relative errors are less than 2.1% for the batteries aged under 35 °C with more than 2000 cycles remaining. The predicted errors are low enough in this case for

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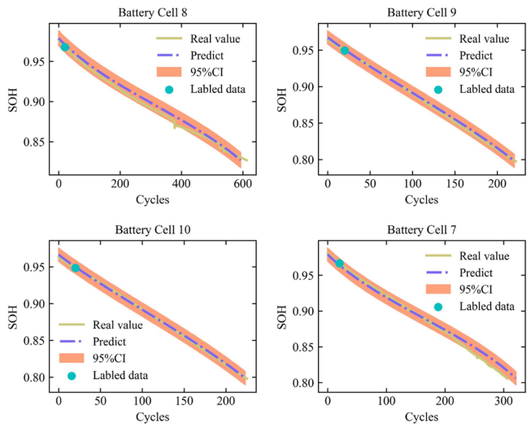


图 12。 TD3中电池的寿命预测结果。

表 IXTD3的容量预测误差

Battery cell	RMSE of capacities	MAE of capacities
Cell 7	0.56%	0.43%
Cell 8	0.53%	0.48%
Cell 9	0.23%	0.19%
Cell 10	0.16%	0.14%

数值结果给出在表 VIII和IX 中，以展示预测寿命和容量曲线的准确性。结果表明，寿命预测误差小于 20 个循环，即使分母小于 300 个循环，相对误差也小于 5.3%。整个预测容量误差的误差也较小。均方根误差不超过 0.56%，平均绝对误差不超过 0.48%，表明预测结果准确。上述预测结果验证了所提出的方法能够在极少数检查点下支持早期寿命预测。

表 VIIITD3的寿命预测误差

Battery cell	Real lifetime	Predicted lifetime	Absolute error	Relative error
Cell 7	296	311	15 cycles	5.21%
Cell 8	595	575	-20 cycles	-3.36%
Cell 9	200	198	-2 cycles	-0.98%
Cell 10	204	202	-2 cycles	-0.97%

进行早期寿命预测。对于所有电池，预测容量曲线的误差都很小。RMSEs 小于 0.81%，MAEs 小于 0.70%，这表明未来容量曲线预测的准确性和可靠性。因此，从上述结果可以看出，即使测试电池与 SD 中的已知电池具有不同的电池类型和不同的老化模式，所提出的仅使用 7 个检查点进行首次再训练和 3 个早期检查点进行自训练的方法，也能预测未来容量衰减曲线和寿命预测。

*2) TD3*中电池的预测： 在本节中，介绍了对第三个测试电池数据集（TD3）的预测结果并进行评估。该数据集的预测方法与TD2相同。首先使用电池单元7对整个NN进行再训练，然后用于预测单元7至10的寿命。对于单元7的预测，首先使用单元10的信息。

这些四个电池的预测结果显示在图 12。由于这种电池的寿命范围较短，窗口大小设置为 30，因此早期退化特征在图中未显示出来。但是，TD3 在 图 1(b) 中显示的整个退化曲线清楚地说明了早期退化。在这种情况下，仅使用一个检查点（第 50 次循环）进行历史容量重建和寿命模型自训练。在 图 12 中显示的预测结果表明，预测的退化曲线接近真实曲线，置信区间满意地收敛于真实值。

V. 结论

退化曲线预测和寿命预测是智能电池管理中的重要且具有挑战性的任务。本文提出了一种基于SSSL的预测框架。首先，从公开数据集中随机选择三节电池用于训练用于电池容量重建和寿命预测的基础模型。从部分曲线中提取了三个历史信息用于容量重建。然后，提出了自训练过程，通过利用伪值来改进模型。此外，只需要少量标记检查点，这符合实际应用中存在少量标记但大量无标签数据的情况。最后，在考虑不同电池类型和/或不同老化特征的多种应用场景下评估了所提出的预测框架。实验结果表明，当基础模型直接用于预测相同电池类型但不同老化特征的电池时，使用三个检查点，平均寿命误差小于23个循环，当使用对应快速变化阶段的另一个检查点时，误差减少到小于8个循环。当该框架用于不同电池类型的寿命预测时，首先使用一个包含七个检查点的电芯重新训练基础模型，然后用于其他电池的预测。结果表明，预测误差小于50个循环，对于寿命超过2000个循环的TD2，在早期300个循环中仅使用三个标记检查点，相对误差小于4.1%。对于寿命较短的TD3，仅使用50个循环时的一个检查点，预测误差不超过20个循环，相对误差不超过5.21%。因此，所提出的方法能够以令人满意的精度和可靠性实现早期未来退化曲线预测。可以看出，所提出的基于SSSL的电池退化预测方法通过容量估计模型重新训练和退化预测模型自学习有助于提高精度。需要注意的是，在其他最先进作品中提出的其他历史信息也可以用于容量估计模型以探索应用范围。在未来的工作中，将考虑温度信息以提高在具有可变温度场景的退化情况下的预测精度。对于具有不同化学成分的电池的退化和寿命预测也将是一个有趣且重要的研究方向。

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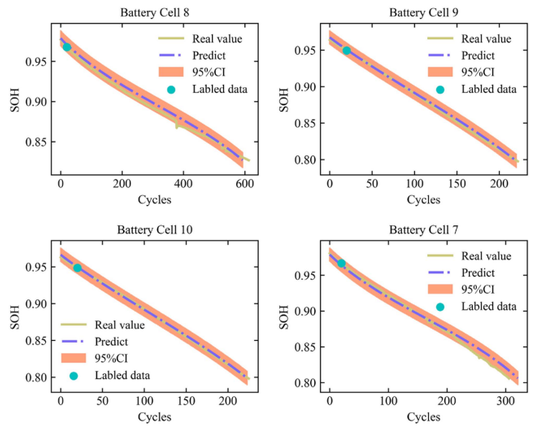


Fig. 12. Lifetime prediction results for batteries in TD3.

TABLE IX
CAPACITY PREDICTED ERRORS OF TD3

Battery cell	RMSE of capacities	MAE of capacities
Cell 7	0.56%	0.43%
Cell 8	0.53%	0.48%
Cell 9	0.23%	0.19%
Cell 10	0.16%	0.14%

The numerical results are given in Tables VIII and IX to show the accuracy of the predicted lifetime and capacity curve. Results show that the lifetime prediction errors are less than 20 cycles, and the relative errors are less than 5.3% even if the denominator is less than 300 cycles. The errors for the whole predicted capacity errors are also small. The RMSEs are no more than 0.56% while the MAEs are no more than 0.48%, indicating accurate prediction results. The above prediction results verify that the proposed method could support early lifetime predictions with very few checkpoints.

TABLE VIII
LIFETIME PREDICTED ERRORS OF TD3

Battery cell	Real lifetime	Predicted lifetime	Absolute error	Relative error
Cell 7	296	311	15 cycles	5.21%
Cell 8	595	575	-20 cycles	-3.36%
Cell 9	200	198	-2 cycles	-0.98%
Cell 10	204	202	-2 cycles	-0.97%

early lifetime prediction. The errors for the predicted capacity curve are small for all the batteries. The RMSEs are less than 0.81% and the MAEs are less than 0.70%, which indicates the accuracy and reliability of the future capacity curve prediction. Therefore, from the results presented above, we can conclude that the proposed method can predict the future capacity fade curve and lifetime prediction with only 7 checkpoints for first retraining and three early checkpoints for self-training even though the testing batteries have different battery types and different aging patterns with the known batteries in SD.

2) Predictions for Batteries in TD3: Finally, the predictions for the third testing battery dataset (TD3) are presented and evaluated in this section. The prediction method for this dataset is the same as that for TD2. The battery cell 7 is first used for the retraining of the whole NN, and then used to predict the lifetime of cells–10. For the prediction of cell 7, the information of cell 10 is first used.

The prediction results for these four batteries are shown in Fig. 12. Since the life ranges of this type of battery are short and the window size is set as 30, the early degradation characteristics are not shown in the figures. But, the whole degradation curves of TD3 shown in Fig. 1(b) clearly illustrate the early degradations. Only one checkpoint (50th cycle) is used for historical capacity reconstruction and lifetime model self-training in this case. The prediction results shown in Fig. 12 indicate that the predicted degradation curves are close to the real curves, with CIs that converge the real values satisfactorily.

V. CONCLUSION

Degradation curve prediction and lifetime prediction are significant, but challenging tasks for smart battery management. This article proposed a novel SSSL-based prediction framework. First, three randomly selected batteries from a public dataset were used to train the base models for battery capacity reconstruction and lifetime prediction. Three HIs from the partial curve were extracted for the capacity reconstruction. Then, the self-training process was proposed to improve the model by taking advantage of the pseudovalues. Moreover, only a few labeled checkpoints were needed, which conforms to the real applications that limited labeled but abundant unlabeled data were available. Finally, the proposed prediction framework was evaluated under different application scenarios, where different battery types and/or different aging profiles are considered. The experimental results show that when the base model was directly used for the predictions of the same battery type, but different aging profiles, the mean lifetime errors were less than 23 cycles using only three checkpoints, and the errors are reduced to less than eight cycles when one more checkpoint, corresponding to the fast-changing stage is used. When the framework was applied for lifetime prediction of different battery types, one cell with seven checkpoints was used to retrain the base models first, and then used for other battery predictions. Results show that the predicted errors are less than 50 cycles, and the relative errors are less than 4.1% for TD2 that have a lifespan of over 2000 cycles with only three labeled checkpoints in the early 300 cycles. The predicted errors for TD3 that have a short life span are no more than 20 cycles with relative errors of no more than 5.21% with only one checkpoint from the 50th cycle. Therefore, the proposed method can achieve early future degradation curve prediction with satisfactory accuracy and reliability. It can be seen that the proposed SSSL-based battery degradation prediction method helps improve the accuracy by capacity estimation model retraining and degradation prediction model self-learning. It should be noted that other HIs proposed

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在其他最先进作品中提出的其他历史信息也可以用于容量估计模型以探索应用范围。在未来的工作中，将考虑温度信息以提高在具有可变温度场景的退化情况下的预测精度。对于具有不同化学成分的电池的退化和寿命预测也将是一个有趣且重要的研究方向。

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in other state-of-the-art works could also be used in the capacity estimation model to explore the application scopes. In future work, the temperature information would be considered to improve the prediction accuracy under degradations with variable temperature scenarios. The degradation and lifetime prediction for batteries with different chemistries will also be an interesting and significant research focus.

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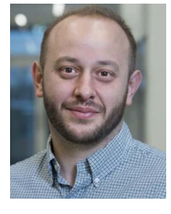
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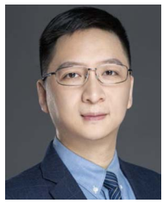
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